Tidal gravity across interrupted optical clock comparisons

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Optical clocks probe gravitational potential through its effect on the passage of time.[1-3] Establishing this capability under interrupted operation requires separating tidal response from clock and transfer variations. Here we analyse independently acquired Shanghai–Wuhan strontium–ytterbium comparisons over a 1,350-km fibre route.[4] Six periods in 2026 provide 20.1 days of retained sample time across a 231-day span. With ten-day calendar baselines, the scale of the predicted tide is 0.88, with a working standard error of 0.10 and a period-resampling 95% acceptance interval of 0.51–1.42. Amplitudes learned from other periods improve the held-out residuals in all six periods. Allowing independent solar-day variations leaves a lunar semidiurnal scale of 0.89, but period resampling no longer excludes zero at the 5% threshold. A simultaneous round-trip record measures sensitivity to one transfer channel. Seasonal solar-tide sensitivity remains more than three orders of magnitude above the predicted differential signal. Together with contemporaneous independent network measurements,[5] these results demonstrate how interrupted clock comparisons test tidal response, while distinguishing accumulated measurement capability from the calibration needed to identify slower gravitational changes.

The gravitational redshift makes a clock's rate a local probe of gravitational potential. Optical-clock development has brought this principle into precision measurement,[6] from laboratory redshift tests[7] to resolving a potential gradient within a millimetre atomic sample.[8] Coherent networks[1,9] and transportable clocks[2] extend the comparison between locations. Static geodetic calibration and time-variable potential require distinct treatment:[10,11] the Moon and Sun, deformation of the Earth and redistribution of mass produce temporal signals superposed on the clocks' intrinsic frequency ratio and systematic shifts.

For an intermittently operated comparison, elapsed calendar time and sensitivity to a tidal component are different quantities. Within-interval variation may constrain a signal even when frequency levels cannot be transferred across every interruption. Conversely, a long calendar span need not constrain a slow signal when clock corrections or instrumental changes remain uncertain. Daily environmental variations create a further distinction between a response to the full predicted tide and the lunar semidiurnal response after allowing solar-day terms.

The contemporaneous European measurements of Stahl and colleagues track tidal potential variations with a clock network and Kalman estimation.[5] Our independently acquired Shanghai–Wuhan comparisons test the response across separated observation periods. The analyses reported here were developed with knowledge of the European work. Our earlier ratio preprint reported a tidal correlation in the summer record.[4] Here we establish a whole-template benchmark, separate lunar and solar-day variation, and examine the different calibration requirements of longer-period sensing. The commissioning status and changed processing of the 2025 observations motivate their separate analysis; they do not enter the common 2026 amplitude. Table 1 distinguishes the overlapping data and statistical targets of earlier and present analyses.

Clocks, link and observation periods

The comparison uses optical lattice clocks based on ^{87}Sr in Shanghai and ^{171}Yb in Wuhan. The associated Sr1 evaluation reports a systematic uncertainty of 9.2×10^{-19}, with chamber thermal characterization and measurements of lattice, density and magnetic shifts.[12] The Yb evaluation reports 1.3×10^{-18}, using an in-vacuum lattice cavity and blackbody shield, and differential comparisons of two clocks.[13] These evaluations describe specified apparatus configurations; they do not provide a complete uncertainty budget for every observation period in the present record.

The 1,350-km route uses commercial fibre, relay stations and bidirectional optical amplification to transfer an optical carrier from Shanghai to Wuhan.[4] Digital phase recording and bias compensation have also been demonstrated on a distinct 2,067-km field link,[14] alongside other long-distance laser-network comparisons.[15] Those results establish technical context. Our transfer check uses the actual simultaneous monitor below rather than importing another route's instability as an error bound.

The principal record contains 1,737,073 retained source measurements in 2,905 non-overlapping 600-s averages, drawn from 49 effective intervals and six observation periods between 7 January and 26

August 2026. Measurements occur on 35 BJT calendar days over a 231-day span. At the nominal one-second gate, the retained source count represents 20.1 days of accumulated sample time. This technical exposure is distinct from elapsed time and from the effective number of independent observations under correlated noise. The periods are January, April, May, June–July, and early and late August. Ratios follow the recorded frequency chain and correction ledger, retaining the measured interval levels relative to one numerical reference (Methods). This coordinate is not a new absolute-ratio determination.

The 2025 records were acquired during clock commissioning, with different frequency conversion and processing and incompletely resolved offsets, according to the experimental team. We analyse September and October separately using one stage intercept or interval intercepts. They are neither joined to each other nor pooled with 2026. This scope follows experimental clarification, not a choice of the year giving the preferred tidal amplitude.

Response across observation periods

We first fit a signed scale A_T multiplying the entire predicted fractional-frequency tide $T(t)$,

$$y(t) = A_T T(t) + \gamma_{b(t)} + \epsilon(t),$$

where $b(t)$ assigns observations to fixed ten-day BJT calendar blocks. Ten blocks contain data. The amplitude and baselines are estimated jointly; measured points are not individually shifted or aligned to the prediction. These baselines are used for the short-period comparison, not for seasonal inference.

The result is $A_T = 0.8836$, with residual-scaled least-squares, delete-day and delete-period standard errors of 0.0882, 0.1002 and 0.0971. We summarize it as 0.88 ± 0.10, retaining the largest standard error. Five-day blocks give 0.886 ± 0.101. Adding an independent linear drift within every occupied block gives 0.878 ± 0.101 for ten days and 0.868 ± 0.112 for five days. The ten-day change is -0.006 ± 0.030 under paired day deletions. These are correlated sensitivity estimates on the same observations.

Figure 1a shows the 48-h window with the largest retained source count, selected by source count rather than tidal agreement. The rule is reproducible but was not prospectively registered before all exploratory analyses. Points retain their measured levels after subtracting one common plotting reference. The displayed template amplitude, 0.830, is estimated from the other five periods; only the prescribed calendar-block baselines are fitted within the held-out period. Three-hour averages aid visualization without bridging gaps.

The actual-time harmonic diagnostic uses the same nuisance projection before and after template subtraction. At the M2 frequency, variance reduction per observation falls from 13.02×10^{-36} to 0.12×10^{-36}, a 99.1% decrease (Fig. 1b). This is a phase-sensitive, in-sample subtraction diagnostic, not an independent discovery test. In leave-period-out comparisons, training amplitudes range from 0.830 to 0.965. Applying them to held-out periods, with their nuisance baselines fitted locally, reduces residual sums of squares in all six periods, by 3.26% when pooled (Fig. 1c). The folds share training data and are not independent experimental replications.

Accumulated sample time provides an exposure measure, but not a universal sensitivity law. Along the actual time-ordered record, eight nominal sample days give whole-template and lunar working errors of 0.114 and 1.025, respectively; at 20.1 days these are 0.100 and 0.282. A fixed independent-bin noise benchmark also shows that lunar separation gains depend on changing phase coverage and nuisance design, beyond exposure alone (Methods, Table 4).

Lunar and solar-day variation

The whole-template model fixes relative tidal amplitudes and phases. Allowing independent sine and cosine terms at exactly 24 h and 12 h within each ten-day block gives $A_T = 0.79 \pm 0.28$, and the M2-frequency diagnostic decreases by 93.8%. Five-day blocks give 0.92 ± 0.38. The daily terms permit instrumental or environmental variations without identifying their origin.

To isolate the lunar semidiurnal response, we fit a common M2+N2 scale while assigning separate scales to the near-12-h and diurnal tidal groups. With the free daily terms retained, $A_L = 0.8909$, with least-squares, delete-day and delete-period standard errors of 0.2344, 0.2824 and 0.1881. The five-day result is 0.89 ± 0.43. Within-block linear drifts give 0.89 ± 0.29 for ten days and 0.82 ± 0.56 for five

days. The lunar response remains compatible with the predicted scale under these specified controls (Fig. 2a).

A common lunar quadrature tests phase consistency without adjusting observation times. The ten-day in-phase and quadrature coefficients are $(0.866, 0.104)$. The unit, in-phase prediction lies within the nominal delete-day regions for both baseline durations (Fig. 2b). The zero point lies inside the five-day region and outside the ten-day region. A separate joint two-parameter test, resampling six periods, gives $p = 0.25$ for zero and $p = 0.72$ for the unit, in-phase point with ten-day baselines. The phase estimates are compatible with the prediction, but the nominal delete-day ellipse alone does not establish a robust exclusion of zero. Three direct-potential alternatives to the solid-tide approximation change the lunar scale by less than 0.005 and the whole-template scale by less than 0.012 (Methods). Separate constituent scales give M2 0.94 ± 0.27 and N2 0.29 ± 1.08; N2 is not independently resolved.

Summer contributes 75.8% of the squared lunar template after projection against the other fitted columns, compared with 45.9% for the whole-template model without daily terms. These are design properties, not physical fractions of the tide. Daily terms also weaken the held-out diagnostic: three of six periods improve, with a pooled reduction of 0.467% (Fig. 1c).

A restricted wild bootstrap over six periods gives two-sided p-values of 0.0313 for zero whole-template response, both without and with ten-day daily terms. The five-day daily-term result is 0.0625. For M2+N2, the corresponding p-values are 0.0625 and 0.0938 for ten- and five-day blocks. With only 64 sign patterns and assumptions about period errors, these tests do not justify a unique high-significance detection claim for the lunar component. Numerical inversion of the ten-day period test gives a 95% acceptance interval of [0.51, 1.42] for the baseline-only whole template and [−1.18, 4.22] for M2+N2. These working sets express the small number of periods and differ from a Gaussian reading of the standard errors. Simulations of the actual error summaries, including their maximum, are reported in Methods. They do not cover arbitrary clock or transfer errors. Actual residual checks expose limitations of a simple homogeneous Gaussian noise model. Additional stress tests with anticorrelation, unequal period scales and heavy tails support the conditional performance of the delete-day phase regions, while exposing undercoverage of delete-period ellipses (Methods). These tests inform uncertainty assessment without identifying the physical noise mechanism.

The measured transfer control

The recorded round-trip monitor tests one part of the transfer system. We use the apparatus-designated B2 channel and pair it with clock measurements without interpolation. A regression bin is retained only when all its original members have unique valid monitor samples. Gross frequency excursions motivate a ±100-Hz screen around the recorded 10-MHz carrier, with unfiltered, ±10-Hz and ±1000-Hz alternatives retained. This screen changes only the monitor comparison's support.

At zero nominal time shift, the ±100-Hz criterion leaves 1,519 bins and 910,041 source members, all in summer 2026. On simultaneous valid second samples, the clock and nominal-optical-normalized round-trip channel have one-hour overlapping Allan deviations of 7.66×10^{-18} and 6.26×10^{-18} (Fig. 3a). On the simultaneous clock–monitor support, the longest evaluated averaging time is 16,384 s, with three contributing runs and a clock Allan deviation of 4.91×10^{-18}. A separate full-2026 diagnostic reaches six hours with two runs from two periods (Methods, Table 6). Neither support establishes stability over the accumulated 20.1 sample days. No assumed factor of two converts this diagnostic into a calibrated one-way bound.

Adding B2 as a regressor changes the whole-template scale by 0.0057 ± 0.0073 without daily terms and 0.0415 ± 0.0345 with them; errors use paired day deletions. Figure 3b shows nominal 95% intervals and threshold sensitivity. Fixed ±5-s pairings are also retained. B2 itself has normalized template projections of 0.074 ± 0.087 without daily terms and 0.54 ± 0.46 with them, using delete-day errors. A small paired change in the clock scale therefore does not demonstrate a zero monitor projection. The result quantifies the tested linear association; clock effects and unmonitored laboratory branches remain possible contributors.

Longer periods and retained levels

For Mf, Mm, Msf and seasonal Sa and Ssa, the 2026 measurements retain their levels and share one global intercept (Fig. 4a). We fit all five quadrature pairs jointly with short-period tides and global daily terms. A single global drift is tested separately. No interval or calendar-block centres are adjusted for these fits.

The no-drift model has insufficient sensitivity to identify the predicted seasonal solar tide. Nominal delete-period 95% upper bounds on free-phase amplitude are $(1.60, 4.32, 2.70, 20.98, 11.11) \times 10^{-17}$ for Mf, Mm, Msf, Sa and Ssa (Fig. 4b). They are conditional statistical bounds: unknown clock corrections and residual calibration changes are not bounded. Adding a global drift expands the Sa and Ssa bounds to approximately 6.0×10^{-15} and 1.1×10^{-15}. Removing May, whose Yb correction is missing, also weakens seasonal constraints.

An independent calculation of the diurnally averaged solar potential gives a semiannual differential response near 2.02×10^{-20}. Similar station latitudes suppress this component. Even the no-drift Ssa upper bound exceeds that prediction by approximately 5,500; the Sa bound is about 75,000 times the current model amplitude. Calibrated levels alone would not close this sensitivity gap. The present sensitivity is far from this predicted signal, and the existing ocean-loading synthesis contains none of the five long-period constituents tested here. More elapsed time alone cannot substitute for calibrated levels and an adequate loading model.

Discussion

The Shanghai–Wuhan comparison supports a common response to the predicted tide across six separated periods from 20.1 nominal days of retained samples. Whole-template scaling, phase-sensitive subtraction and internal transfer of the amplitude support this interpretation within the stated models. Allowing solar-day terms retains a lunar response compatible with the prediction; joint phase tests and residual diagnostics limit component-specific attribution. Accumulated sample time quantifies the observations acquired, while phase coverage, calibration and noise structure determine their sensitivity. The measured monitor tests a specified transfer association while leaving clock and unmonitored-path contributions within the interpretation boundary.

The earlier ratio result and this analysis use overlapping observations for different targets (Table 1). The reported $0.87(15)$ and 5.7σ in the earlier, narrower model cannot be transferred to the present lunar estimate. More general nuisance and noise tests do not by themselves invalidate that earlier statement. The commissioning data are now kept separate because experimental configuration, not calendar length, defines which measurements can be combined. Unknown corrections and the retained-level plotting coordinate prevent direct interpretation of Fig. 4 as the calibrated absolute-ratio coordinate of the earlier summer analysis.

The European work establishes tidal tracking with a network, Kalman estimation and same-site controls.[5] Our independent comparison addresses consistent response over interrupted operation, separation from solar-day variation and the limits of retained-level inference. This is complementary experimental evidence, without claiming exclusive clock species, geographical scale or analysis techniques. Neither the whole-template scale nor this component decomposition independently estimates geophysical response parameters.

Dynamic chronometric sensing thus has more than one calibration timescale. Semidiurnal variations can be tested within observation blocks while response consistency is examined between periods. Seasonal sensing requires those relative levels themselves to remain calibrated. Improved clock and local-transfer monitoring would strengthen both short-period attribution and the search for slower potential changes.

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Table 1

Table 1 | Related tidal estimates and their statistical targets. The estimates share observations and are not combined. Present common amplitudes use 2026 only, with 2025 commissioning records analysed separately. Values in parentheses or after ± denote the error scale specified in the last column; they are not a common calibrated confidence level.

Analysis	Observations and averaging	Tidal target and nuisance model	Scale	Error definition
Hou et al.[4]	Summer record; 16 intervals; 1,200-s triangular windows	Whole predicted tide; interval-centred, precision-weighted estimates; window roles distinguished below	0.87(15)	Reported weighted statistical error; reported 5.7σ under that analysis
Present 2026 reconstruction	2,905 non-overlapping 600-s means	Whole predicted tide; ten-day calendar baselines	0.88 ± 0.10	max(LS, delete-day, delete-period SE)
Present daily-term control	Same 2,905 means	Whole predicted tide; ten-day baselines and 24-h/12-h terms	0.79 ± 0.28	Same standard-error summary
Present lunar decomposition	Same 2,905 means	M2+N2; independent other tide groups and ten-day daily terms	0.89 ± 0.28	Same standard-error summary

The archived calculation uses 600-s-stride, 1,200-s triangular windows for amplitudes, but non-overlapping windows for standard errors and correlations, contrary to the common support stated in Hou et al.'s Table 8.[4] Using non-overlapping windows for amplitudes as well changes the precision-weighted scale from 0.87303 to 0.87756, with the same conditional standard error of 0.15193. This comparison does not recalibrate the reported significance.

On identical current summer observations (1,642 averages), ten-day baselines and common-amplitude least squares, the published and current templates give scales of 0.752 and 0.693. Historical-window comparisons also depend on precision weighting versus a common-amplitude fit. These controlled comparisons do not establish which template is physically more accurate or fully disentangle changes in selection, frequency conversion, timing and averaging. Extending from summer to all six periods additionally changes observing support; agreement with the earlier aggregate is not independent validation.

Figure legends

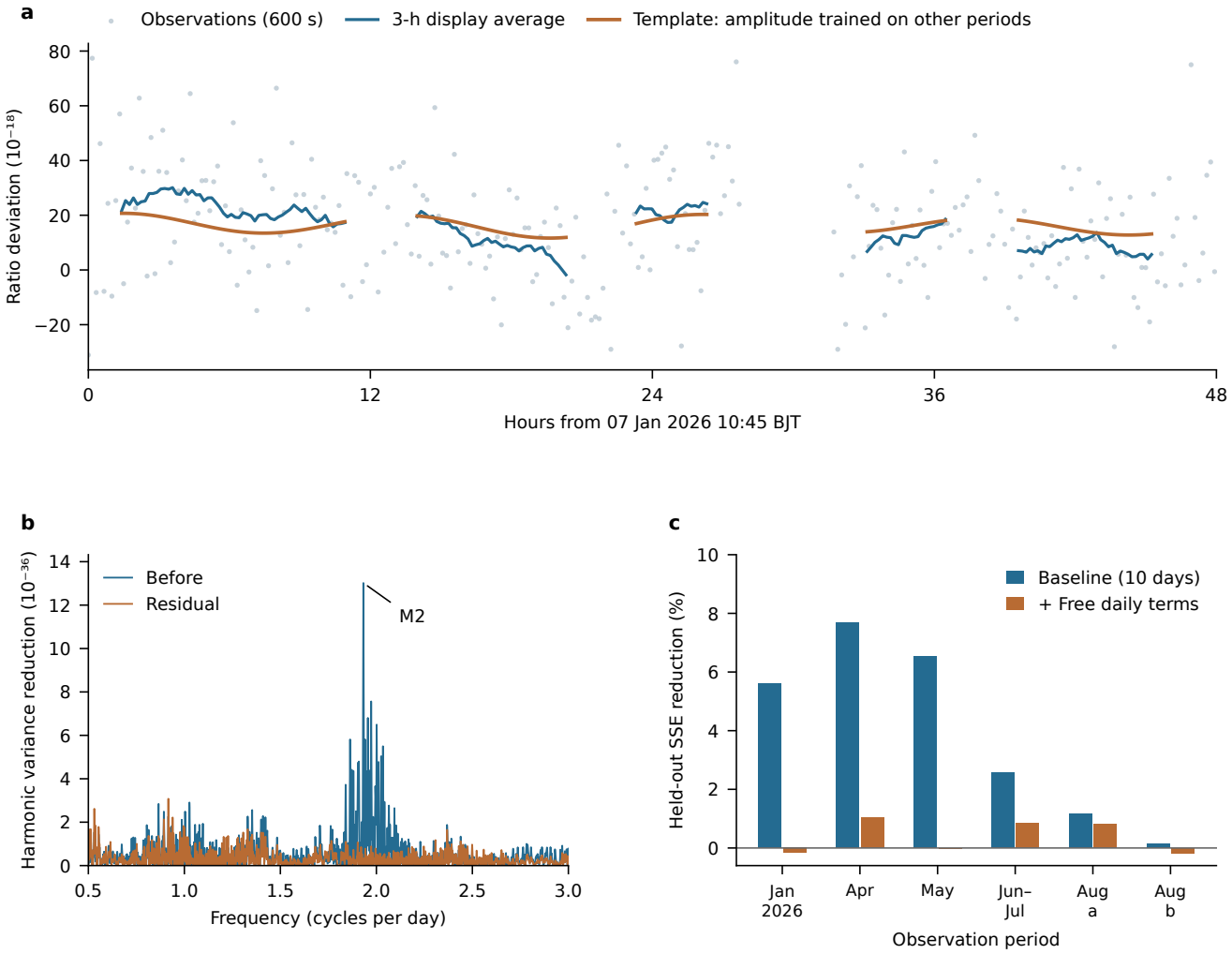


Fig. 1 | A common tidal-template response across interrupted observations. a, The 48-h window containing the largest number of retained source members (158,072 in 264 bins), selected without reference to fit quality. Points are 600-s averages after subtraction of one common reference used for the whole record. The blue curve averages 18 consecutive bins within uninterrupted supports for display. The orange curve applies the same display average to the template, with its amplitude trained on the other five periods and its calendar-block baselines fitted in the held-out period. No individual segment offsets are subtracted from the points. **b,** Actual-time harmonic scans of all 2,905 bins before and after whole-template subtraction, with the same ten-day baseline projection. The ordinate is harmonic variance reduction per observation, not a physical power spectral density or a significance threshold. **c,** Changes in held-out-period residual sums of squares when the amplitude is learned from the other periods. Blue and orange denote the baseline-only and free-daily-term designs. The held-out nuisance terms are fitted locally; the six folds share training data. All time-series points are repeated observations of the same apparatus, not independent experiments.

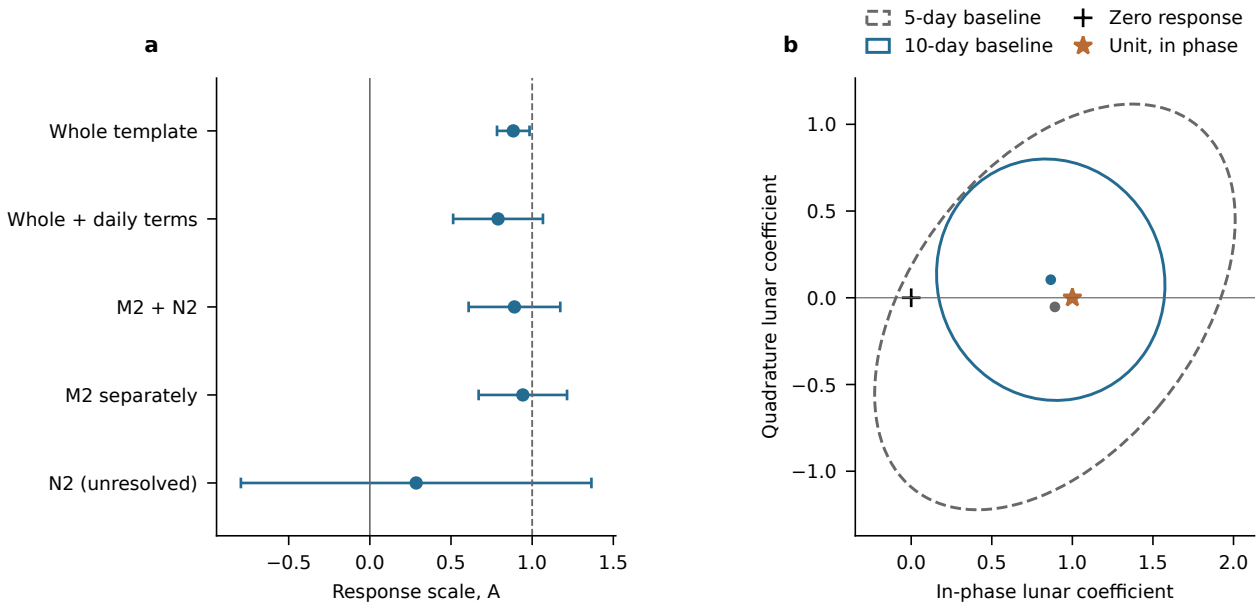


Fig. 2 | Lunar response after allowing solar-day variation. **a,** Scales estimated from the same 2,905 averages. All rows except the first include ten-day 24-h and 12-h nuisance terms. M2 and N2 are estimated jointly with independent scales in the last two rows. Bars show one standard-error summary, max(LS, delete-day, delete-period); the vertical references mark zero and unit response. The rows are correlated estimates of different targets. **b,** In-phase and quadrature coefficients of a common lunar response for five-day and ten-day baselines. Curves are nominal 95% delete-day working-model regions for the two coefficients; points mark their estimates. The cross and star denote zero and the unit, in-phase prediction. Zero is inside the five-day region and outside the ten-day region; the unit, in-phase point lies within both. No observation times or phases are adjusted. For the same two-parameter target, the six-period wild test gives ten-day $p = 0.25$ at zero and $p = 0.72$ at the unit point (Methods, Table 5); it uses different covariance estimation and calibration from these ellipses. Neither procedure corrects for exploratory model selection or arbitrary noise.

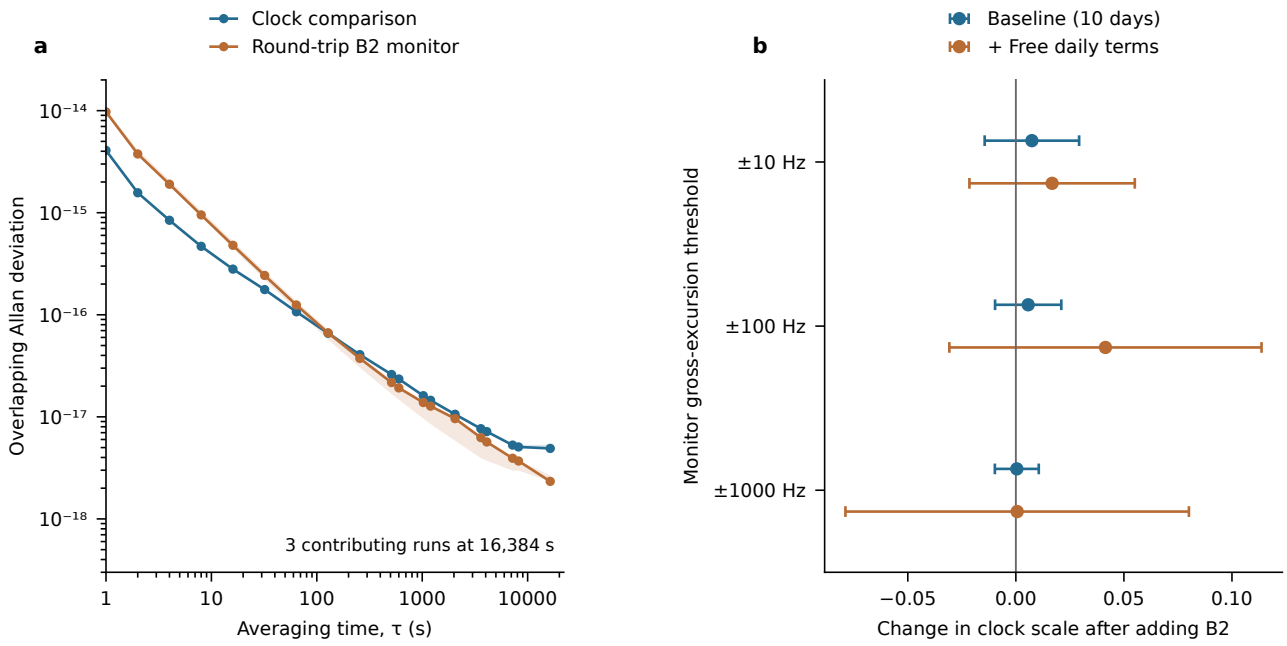


Fig. 3 | Scope of the measured round-trip transfer control. **a,** Overlapping Allan deviations of the clock comparison and predesignated B2 monitor on simultaneous valid second samples. Solid curves use the ±100-Hz gross-excursion criterion (913,692 common seconds before averaging-time run requirements). Shading spans the ±10-Hz and ±1000-Hz sensitivity results as well as the main curve; it is not a confidence band. No differences cross gaps, and contributing runs must span at least four averaging times. At 16,384 s, only three runs contribute 268,673 overlapping differences, which are not independent replicates. The monitor is normalized by the nominal optical carrier without an assumed one-way conversion. **b,** Paired changes in the clock's whole-template scale on adding B2, at zero nominal pairing shift. Thresholds leave 1,476, 1,519 and 1,520 complete 600-s bins. Error bars are nominal 95% delete-day intervals for the paired changes, with 19 degrees of freedom; they are not interval differences computed from independent fits. Baseline-only and free-daily-term designs use the same support within each threshold. The control covers the recorded channel and time overlap, including neither unmonitored laboratory branches nor the complete year.

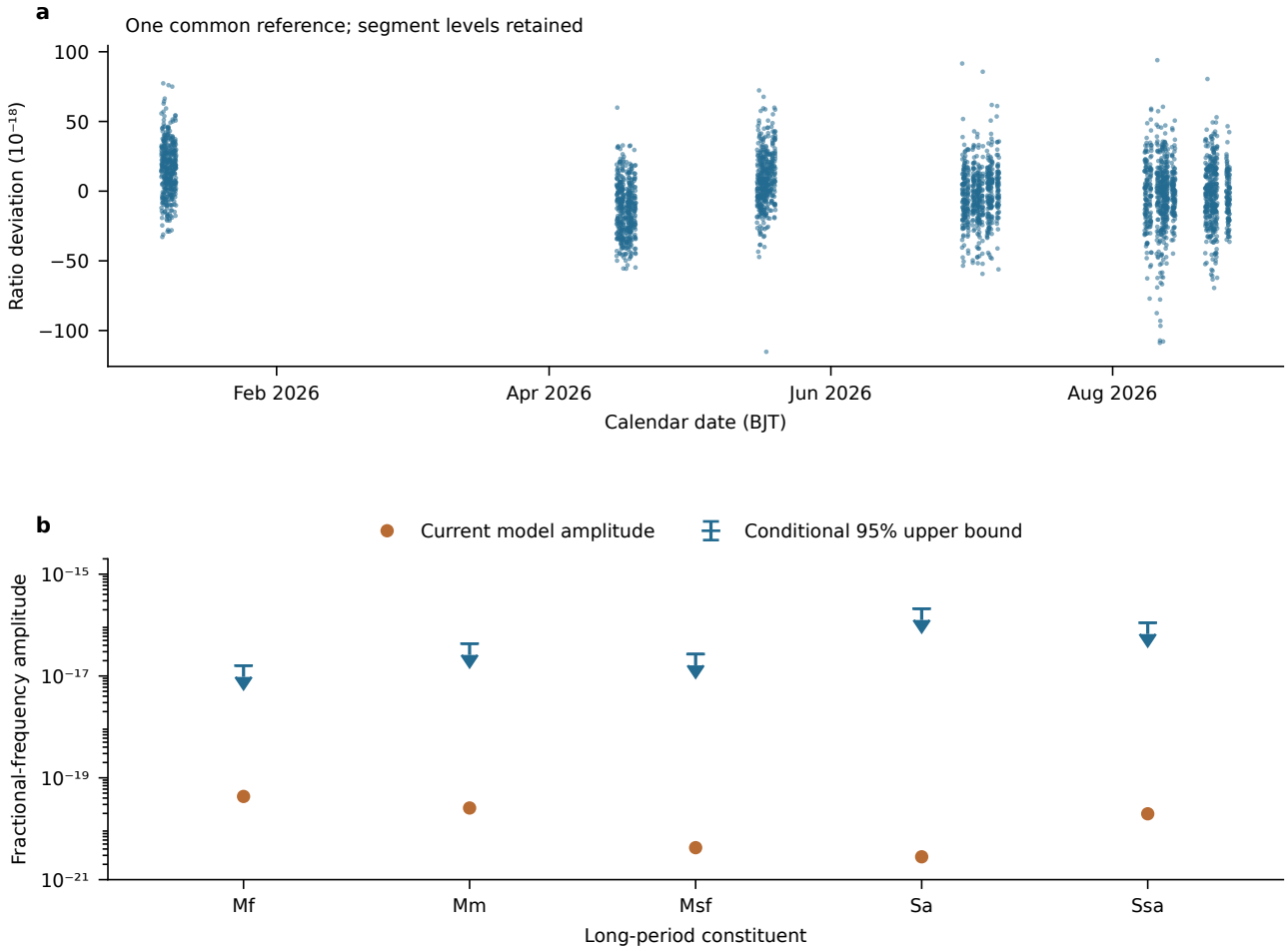


Fig. 4 | Preserved frequency levels and long-period sensitivity. a, All 2,905 reconstructed 2026 ratio averages, with one common plotting reference subtracted. No fitted segment or period centres are removed. Points retain the actual gaps and period-to-period level differences. **b,** Nominal 95% upper bounds on free-phase amplitudes from the 2026-only, single-intercept model with joint short-period and global daily terms, without a global drift. Caps indicate upper bounds and arrows indicate the permitted lower-amplitude direction; arrow lengths are graphical. Bounds use the largest eigenvalue of each two-coefficient delete-period covariance and an approximate two-dimensional region with five denominator degrees of freedom. They are individual bounds, conditional on the stated calibration and noise assumptions. Orange points give amplitudes from the current theoretical decomposition, not measured signals. The five-constituent family bounds and drift sensitivity are specified in Methods. Unknown calibration terms and incomplete long-period loading prevent these bounds from being interpreted as unconditional gravitational limits.

Methods

Coordinate and frequency conversion

The principal inference uses 2026; the separately recorded 2025 conversion is retained for commissioning-stage diagnostics.[4] Let $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The 2025 chain gives

$$f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \quad \alpha = \frac{10.93629485}{25N_{1397}}, \quad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight Shanghai reference units include seven in the transmitted carrier and one added at Wuhan. The Wuhan repetition frequency is $f_{\mathrm{Yb}}/(2N_{1156})$, its reference is $f_{\mathrm{Yb}}/(40N_{1156})$, and its comb offset is 26 reference units. For recorded frequency q in nominal hertz realized by that Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

We use $d = 2$ and the stated α in 2025, versus $d = 1$, $\alpha = 0$ in 2026. The respective counter values are near 66 and 33.6 MHz; the former represents a 132-MHz beat before division. The counter-reference convention is an experimental assumption, not implied by a 10-MHz label. The electronic divider is restored once, independently of any round-trip correction.

For signed Sr and Yb shifts a and b, $R = R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$, with inherited $\Delta G = -3.116\times 10^{-15}$. We form $y = (R/R_{\mathrm{ref}} - 1)10^{18}$ using $R_{\mathrm{ref}} = 1.2075070393433377166$ and Decimal anchors, avoiding subtraction of full ratios in float64. Positive y means faster Wuhan Yb relative to Shanghai Sr. For positive Newtonian potential U, the rate contribution is $-[U_{\mathrm{WUH}} - U_{\mathrm{SHA}}]/c^2$. These conventions do not establish a new absolute ratio or static potential.

The eight periods are September/October 2025, January/April/May 2026, June–July, and early/late August. In 10^{-18} units, Sr density corrections are [−0.43,−0.41,−0.30,−0.33,−0.25,−0.24,−0.33,−0.29], the first six Zeeman corrections are [−178.50,−178.87,−179.14,−180.19,−180.18,−178.57], and the background-gas term is −0.67. August Zeeman and all blackbody corrections are recorded as online. The first five AC entries are missing; the final three are [7.44,8.93,9.03]. Positive 37.2-mHz feedback applies to the first three periods at the assumed 698-nm optical node, giving 86.6119×10^{-18}—larger than the approximately 10×10^{-18} tide. Printed Yb shifts for original groups 1–9 are primary, with component sums tested separately; May Yb is missing. Computational zeros for unknown entries are not measured zeros. Missing terms have no certified bound. The ledger is frozen for these refits, but neither its precedence over all earlier residual inspection nor within-block constancy is established. Baselines absorb constants; time-dependent shifts can project onto the tide.

Commissioning records and corrections

Reported commissioning, processing changes and unresolved offsets motivate separate September and October 2025 stage- or interval-intercept analyses, without shared amplitudes across months or with 2026. This scope was clarified during reanalysis.

An older uncompensated Yb shift near -1.1×10^{-17} may overlap the updated totals -9.7 and -9.5×10^{-18}, which include a lattice entry of -8×10^{-18}. Inclusion is unverified; we do not add it twice. An additional δb multiplies R by $(1+b)/(1+b+\delta b)$, so $\delta b = -1.1\times 10^{-17}$ would increase R by approximately that fractional magnitude. Additive-shift invariance of intercept designs does not verify this multiplicative correction or its provenance. September and October stage fits give 0.87 ± 0.67 and 1.57 ± 0.26 (LS only), with larger errors after daily controls. Only two days per month preclude establishing correlated-noise coverage; these are commissioning diagnostics.

Observation selection and averaging

The corrected intervals select 1,966,912 unique source rows from 42 files, retaining fractional-second BJT labels. Windows include the first PDF second and exclude the second following the last. Rows are time/file/row sorted and deduplicated by source identity. Finite offsets below 1 MHz in magnitude and

strictly within 10 Hz of the plausible-sample median leave 1,966,908 rows. The three-row difference from the experimental count retains boundary provenance.

Within each interval, 300-s block means are compared with a reflected-boundary, 13-block running median. Residual median absolute deviation divided by 0.6744897501960817 sets the screening scale. Blocks exceeding six scales and their immediate neighbours are excluded, removing 11,131 rows. Local residuals select samples; they do not replace retained beat values, and no fitted tide residual enters selection. Non-overlapping 600-s means require at least 300 members: 3,266 means contain 1,952,992 members. Restricting to 2026 leaves 2,905 means and 1,737,073 members; September/October 2025 retain 160/201 means. Theory columns use identical members. Baseline indicators use mean bin times, retaining complete boundary-crossing bins; observations are not interpolated. Day deletions use mean-bin BJT days, not every raw sample of that civil day: the parent record includes 16 midnight-crossing bins. Retained 2026 samples occupy every local hour, with 45.6% between 06:00 and 18:00. Table 3 lists ten-day support. No block crosses the year-convention change, and within-block drift tests identify no specific hardware event.

Tidal prediction and component design

The common prediction combines solid-Earth and ocean-loading series on a 90-s grid. Pysolid displacements under the recorded IERS 2010 implementation[16] are converted using

$$s(t) = -\frac{1 + k_2 - h_2}{h_2}\,\frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m\,s^{-2}}$. This displacement approximation is not a complete degree-dependent potential calculation. Loading uses 18 constituents from grids locally documented as FES2022b.[17] The FES2014 paper describes an earlier atlas, not validation of FES2022.[18] FES2022 was funded by CNES, produced by LEGOS, NOVELTIS and CLS, and made freely available by AVISO. Convolution uses PREM load-Love Green functions containing $1+k'_n-h'_n$; the Han–Wahr table[19] through degree 696 is extended to 40000, tapering the final 30%, with the recorded centre-of-solid-Earth degree-one convention. Direct water potential, induced potential and displacement are included. Member-matched solid, loading and total demeaned RMS values are $(3.245, 1.507, 4.410) \times 10^{-18}$. Loading/total RMS is 34.2%, with solid–loading correlation 0.680; variances are not additive. Atmospheric loading is neither modelled nor bounded. Free daily terms control empirical variation but do not validate an atmosphere-loading field, nor can local pressure differences do so.

The 360,001 theory samples are projected onto quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant (rank 27), retaining the remainder. BJT converts to UTC once, with the saved FES convention and TT−UTC=69.184 s. Whole-template fits scale the complete prediction; component fits subtract the remainder once. In 10^{-18} units,

$$y - F = A_L C_L + a_{12} C_{12} + a_D C_D + Z_b \gamma + Z_{24,12}\beta + \epsilon.$$

C_L, C_{12} and C_D contain M2+N2, S2+K2 and the four diurnal constituents; F contains remaining theory, including long-period terms. The daily design has sine/cosine columns at one and two cycles per day in each occupied block. The ten-day lunar design has 53 parameters and full rank. Blocks follow $\lfloor t/(10 \times 86400) \rfloor$ on the BJT epoch coordinate; five-day blocks follow the analogous rule. S2+K2 is not interpreted as resolved from exact 12-h terms. Drift tests add a within-block linear column, centred on observed mean time, on identical data; paired deletions estimate changes. These short-period baselines/slopes never enter seasonal fits.

Twenty-day, thirty-day and per-period controls give lunar scales 0.867 ± 0.246, 0.796 ± 0.202 and 0.761 ± 0.205 from the same sample time. Thirty-day blocks share a baseline/daily pattern across August periods; per-period models separate them. Their scalar wild $p = 0.03125$ does not motivate replacing the weaker five-/ten-day primary controls (joint tests: Table 5).

Three sensitivities replace only the solid-Earth series with DE430 geometric potentials:[20] degree two with response $1+k_2-h_2$; degrees two/three with constant responses; and degrees two/three with nominal order-dependent real k_{nm} from IERS Table 6.3.[16] We use $h_3 = 0.292$, IAU 2006/2000A rotation, UT1≈UTC and zero polar motion. These first three alternatives hold the response coefficients constant in frequency. Whole-template scales become 0.876, 0.872 and 0.872; lunar scales become 0.895,

0.887 and 0.887. For the degree-two constant-response alternative, substituting Hou et al. coordinates[4] (Wuhan 30°31′ N, 114°19′ E, 30 m; Shanghai 31°14′ N, 121°29′ E, 5 m), with the same potential algorithm and loading, changes A_T by +0.0066 and A_L by +0.0046; solid-template difference RMS is 0.043×10^{-18}. Neither coordinate set is certified at the clock transition points; height references also require confirmation.

A separate finite-basis sensitivity applies nominal frequency-dependent $\gamma(\omega) = 1+k(\omega)-h(\omega)$ from the IERS resonance and anelastic approximations[16] implemented in pyTMD 3.0.9. We project the constant-response degree-two DE430 series onto the same 13 constituents, transform each pair by γ/γ_0, and retain the constant and residual from each projection and the original loading series. Here $\gamma_0 = 1 + 0.3010 - 0.6078$. Both projections use equal weights; transformed series are evaluated on the original full grid and averaged over the unchanged source members. For $C\cos\theta + S\sin\theta$ and $\gamma/\gamma_0 = a+ib$, the transformed coefficients are $(aC + bS, aS - bC)$. Relative to this same DE430 constant control, real-only and complex responses change the baseline-only whole-template scale by -0.00442 and -0.00434, respectively; changes in daily-controlled A_L are below 0.00008. Trimming 30 days from each end of the theoretical projection window changes these scales by less than 0.000007. The observations and reported primary estimates are unchanged; alternative estimates are retained as sensitivities. This nominal finite-basis check does not reproduce the full effective IERS response, separate nodal-sideband responses, latitude dependence or complete ocean feedback; its small shifts do not bound all theory errors.

Estimation, correlation and validation

Column-normalized SVD retains residual-scaled LS, delete-day and delete-period covariances. Long-period analyses additionally retain elapsed-time Bartlett HAC sensitivities at one, three and ten days. Jackknife covariance[21] is $C_{\mathrm{JK}} = (G-1)G^{-1}\sum_g(\hat{\theta}_{-g} - \bar{\theta})(\hat{\theta}_{-g} - \bar{\theta})^T$. Deletion-group independence remains an assumption. Unsupported nuisance columns are removed; target/rank loss is recorded. The maximum LS/day/period SE is a scalar summary, not a joint covariance. Two-coefficient regions use $(b-\hat{b})^T C^{-1}(b-\hat{b}) \le 2F_{2,\nu}(0.95)$, with $\nu = G-1$ for jackknife covariance and $n-p$ for LS. Figure 2b uses delete-day covariance with 34 denominator degrees of freedom.

Restricted wild tests[22] fit each null and enumerate 64 period-wise sign patterns, studentizing with delete-period covariance. Scalar tests use bootstrap-t; joint phase tests use both coefficients and their covariance (Table 5). Period independence and error symmetry are working assumptions; nuisance estimation prevents exact randomization inference. The minimum two-sided p-value is 0.03125. Exploratory choices are not multiplicity-adjusted. Scalar inversion partitions the real line at polynomial roots, including endpoints and isolated-point checks. The ten-day daily-term whole-template 95% set is $[-0.541, -0.294] \cup [0.241, 0.367] \cup [0.385, 2.568]$, not its convex hull. Independent QR and covariance checks verify implementation agreement, not experimental coverage.

Fixed-design simulations retain actual times: 1,024 per design/scenario, seed 202611052. Errors are Gaussian iid variance 500 in working units; white SD 10 plus OU SD 20 with timescale 1 h, 12 h or 3 days; or white SD 10 plus independent period constants SD 30. Constants may be absorbed by nuisance columns and do not constitute a general drift test. Table 2 reports ten-day whole-template and lunar performance. $\hat{A} \pm u_{\max}$ is a one-SE summary; normal and t_5 multiples define separate nominal 95% intervals. True-value wild acceptance is distinct from numerical inversion. The historical 124/200 injection coverage concerned one SE, not 95%, and cannot certify extreme-tail significance.

A Gaussian white-plus-OU diagnostic averages exponential covariance over actual member pairs, with white-bin variance proportional to $600/n_i$. ML/REML profile a common scale over iid and 70 combinations of doubling timescales 10–5120 min and mixing fractions 0.1, 0.3, 0.5, 0.7, 0.9, 0.97 and 0.995. Whitened residuals show lag-one correlations −0.13 to −0.17, period RMS ratios 1.37–1.38 and kurtosis 4.82–5.12. None of 4,096 conditional simulations per selected model exceeds the observed maximum absolute lag-1–12 correlation, RMS ratio or kurtosis. They retain mean projection but do not reselect covariance parameters; these are not selection-adjusted probabilities. This noise family is inadequate for those diagnostics, and its likelihood significance is not used for detection.

Three further stress scenarios use variance-normalized Student-t7 or t5 innovations, negative MA(1) correlation −0.15, and fixed period scales from whole-template residuals. A third mixes t7 MA and member-averaged 12-h OU at variance fractions 0.9/0.1, adjusting MA correlation to approximately preserve lag one. MA memory restarts across interval changes or gaps ≥900 s. White-bin variance

scales as $600/n_i$; period scales are $[18.518, 17.888, 19.313, 19.170, 24.699, 21.638] \times 10^{-18}$. Each has 4,096 simulations (seed 202611056), without re-estimating these empirical parameters or reselecting samples. Delete-day joint-phase regions cover 96.8–98.2%, versus 85.8–86.9% for delete-period ellipses and 96.8–96.9% wild acceptance. Joint wild power for unit response is 19.9–27.6%, versus 99.0–99.8% for the whole template without daily terms. These exploratory conditional tests neither identify instrument noise nor certify arbitrary systematics or model selection.

Exposure and conditional sensitivity

Chronological prefixes first reach 2, 4, 8, 12 and 16 nominal sample days, plus full support, retaining complete bins (Table 4). Thresholds preceded this calculation but followed full-data exploration. Nested prefixes change phase coverage, period composition and nuisance columns; they are not independent replications or sequential tests. LS, day and period errors are separate; fewer than three deletion groups are marked unavailable. Precision need not improve monotonically.

A conditional technical benchmark uses the measured summer 600-s clock Allan deviation, $\sigma_{600} = 2.3476\times10^{-17}$, as a white-frequency reference. Independent bin variances $v_i = \sigma_{600}^2(600/n_i)$ propagate through the unchanged OLS target operator O as $O\,\mathrm{diag}(v_i)O^T$. This does not reweight observations or certify independence and equal noise across periods. The full-support benchmark errors are 0.101 and 0.268 for whole-template and lunar scales. At four nominal days, the lunar benchmark error is 4.33 times that obtained by scaling the full-support error as $T^{-1/2}$, reflecting the different sampling and nuisance geometry. No Allan extrapolation to 20 days or universal detection time is inferred.

Spectral and held-out diagnostics

Harmonic scans generalize floating-mean least squares[23] using the nuisance projector $M_Z = I - ZZ^+$. At each frequency, the two-quadrature fit to $M_Z y$ gives residual-sum-of-squares reduction divided by sample size. Actual bin-mean times scan 0.25–4 cycles/day at $1/(5\Delta t)$ spacing, including specified constituents. Subtracting $\widehat{A}_T M_Z T$ leaves the same projection. Gaps remain intact. This is neither a physical PSD nor a calibrated frequency-search significance test.

For leave-period-out diagnostics, five periods train the amplitude. The test response and template are projected against locally fitted nuisance columns before comparing residual sums of squares. The test amplitude is fixed; training observations overlap across folds.

Fibre monitor and stability

The 29 monitor files use the experimental B2 designation. Their nominal one-second BJT axes start at the floored first timestamp and advance by row index, matched to corresponding clock axes; repeated or nonfinite monitor seconds are excluded. Pairings at −5, 0 and +5 s test that convention without certifying synchronization or latency. All original members of a retained bin must match. Zero-shift unfiltered support has 915,039 samples, including 1,347 B2 deviations beyond 100 Hz. The post-inspection ±100-Hz excursion screen retains ±10/±1000-Hz alternatives and was not selected for tidal agreement.

We compare $y = A_T T + Z\eta + \epsilon$ with $y = A'_T T + Z\eta' + \lambda B2 + \epsilon'$. Paired day deletions estimate $A'_T - A_T$ uncertainty, retaining B2's own template projection. Frequency deviations use nominal optical carrier 193399200000000 Hz for a round-trip diagnostic; neither division by two nor 10-MHz RF normalization supplies a one-way optical error bound.

Overlapping Allan deviations[24] compare adjacent τ-means on contiguous common seconds, with no term across gaps and no run shorter than 4τ. Term counts are retained. Summer results are not extrapolated to other periods; timestamp checks do not certify hardware synchronization. No same-site null comparison is available.

For the full retained 2026 clock membership, nominal-second runs additionally split at file and effective-interval boundaries. The primary minimum length is 4τ; a separate 2τ rule is exploratory (Table 6). At six hours, two runs from two periods give 4.35×10^{-18} before and 2.16×10^{-18} after subtraction of the fixed unit-amplitude prediction on identical support, without fitted amplitudes, phases, drifts or offsets. The latter is a model-conditional diagnostic, not an independent tidal test or instrument-noise bound. Eligible runs change with τ, so the curve is not solely an integration-gain law. The exploratory twelve-hour estimate has only seven overlapping differences. Overlap counts are not independent degrees of

freedom, and no calibrated confidence interval is assigned. This full-record diagnostic does not replace the summer common-support benchmark used in the exposure analysis.

Long-period amplitudes and upper bounds

Long-period models jointly fit five quadrature pairs, short-period group scales and global daily terms on 2026 data with one intercept. Alternatives add one global drift, use component-sum corrections or omit May. No segment, period or month intercepts are used; all 2025 measurements are excluded.

For a pair $\hat{b}$ with covariance C and denominator degrees of freedom ν, a conservative radial bound from the nominal ellipse is

$$U = \|\hat{b}\| + \sqrt{2F_{2,\nu}(1-\alpha)\lambda_{\max}(C)}.$$

We use $\alpha = 0.05$ for an individual bound and 0.01 for the five-target Bonferroni family. With six period clusters, $\nu = 5$ is a working approximation. The no-drift family upper bounds for Mf, Mm, Msf, Sa and Ssa are $(2.30, 6.18, 3.75, 30.38, 15.28) \times 10^{-17}$. Omitting May leaves five periods and gives individual Sa and Ssa bounds of approximately 1.37×10^{-15} and 9.36×10^{-16}. Bounds are conditional on the calibration coordinate; missing early Sr AC corrections and other unbounded changes prevent an unconditional gravitational limit.

The independent solar check averages the degree-two potential over hour angle. With P_2 the second Legendre polynomial, it evaluates $-(1+k_2-h_2)GM_\odot[r_W^2 P_2(\sin\varphi_W) - r_S^2 P_2(\sin\varphi_S)]P_2(\sin\delta)/(r_\odot^3 c^2)$ using DE430[20] and the archived theory-input coordinates (Wuhan: 30.529° N, 114.357° E, 50 m; Shanghai: 31.130° N, 121.570° E, 20 m), converted to geocentric radii and latitudes for this calculation. This checks the small Sa/Ssa scale without fitting observations. It does not provide a complete seasonal-loading model. The existing 18-constituent loading series contains none of Mf, Mm, Msf, Sa or Ssa.

Computational reproducibility

Independent acquisition is reported by the experimental team; new template, spectral and component analyses followed access to the European work and critique. Route, species and geography are not claimed as exclusive categories.

Versioned files retain source identities, exclusions, designs, covariances, seeds and figure inputs. An independent parser read all 1,737,073 retained 2026 records from 34 contributing original files, verifying hashes, row counts and identities. Direct 80-digit Decimal conversion and member averaging reproduce both ratio-coordinate routes within 8.2×10^{-28} fractional frequency. This verifies the frozen data path, not sample selection or experimental corrections. Independent QR/covariance checks agree. Principal calculations use Python 3.12.3, NumPy 2.5.4 and SciPy 1.18.1; astronomy uses NumPy 2.5.3 and pyTMD 3.0.9; figures use Matplotlib.

Data availability

Versioned manifests, derived series, exclusions and figure inputs are retained locally. Authors must specify access and restrictions before submission; no public deposit is claimed.

Code availability

Versioned analysis, validation and plotting code includes source hashes and execution records. Authors must confirm access for editors, referees and readers; no public code release is claimed.

Table 2 | Conditional simulation performance. T: ten-day whole template without daily terms; L: M2+N2 with daily terms. Each Gaussian scenario uses 1,024 simulations; each Student/MA scenario uses 4,096 with fixed unequal period scales. **a,** Bias and mean scale standard errors. **b,** Nominal 95% coverage (%): individual LS/day/period t intervals, $1.96u_{\max}$, and wild true-value acceptance. These are simulated responses, not observations or universal coverage guarantees.

a. Bias and standard errors

Target / noise	Bias	LS SE	Day SE	Period SE
T / iid	-0.000	0.096	0.096	0.091
T / OU 1h	-0.007	0.094	0.264	0.255
T / OU 12h	-0.002	0.084	0.247	0.226
T / OU 3d	+0.003	0.064	0.146	0.134
T / period constants	+0.001	0.043	0.043	0.041
T / t7 MA	+0.002	0.088	0.074	0.071
T / t5 MA	+0.001	0.088	0.074	0.071
T / t7 MA + OU	+0.001	0.087	0.098	0.090
L / iid	-0.001	0.255	0.298	0.270
L / OU 1h	-0.002	0.238	0.799	0.714
L / OU 12h	+0.031	0.207	0.837	0.631
L / OU 3d	+0.004	0.164	0.594	0.359
L / period constants	+0.004	0.114	0.134	0.122
L / t7 MA	+0.004	0.236	0.244	0.213
L / t5 MA	+0.003	0.236	0.243	0.213
L / t7 MA + OU	+0.005	0.231	0.336	0.262

b. Interval coverage (%)

Target / noise	LS	Day	Period	Max-SE	Wild
T / iid	95.0	94.8	94.2	96.7	96.5
T / OU 1h	49.9	94.1	93.7	94.5	96.5
T / OU 12h	49.7	95.6	94.1	96.1	97.4
T / OU 3d	60.9	96.4	94.0	95.9	96.9
T / period constants	94.7	93.9	93.8	96.0	96.0
T / t7 MA	97.9	94.6	94.6	98.3	96.5
T / t5 MA	98.4	95.2	94.7	98.9	96.7
T / t7 MA + OU	93.7	96.4	94.9	97.2	97.0
L / iid	95.2	96.8	94.4	98.3	96.9
L / OU 1h	50.2	96.3	93.1	96.9	96.7
L / OU 12h	49.9	99.0	92.9	98.8	96.5
L / OU 3d	67.0	99.8	93.5	99.8	95.8
L / period constants	96.2	97.5	94.1	98.3	97.5
L / t7 MA	96.9	96.7	92.7	98.2	96.6
L / t5 MA	97.1	97.0	92.4	98.3	96.7
L / t7 MA + OU	93.1	98.5	93.0	98.5	96.9

Table 3 | Ten-day block support in 2026. Assignment uses the mean bin time; boundary-crossing source members remain intact. The start is inclusive and the end exclusive. All dates are BJT in 2026. Block duration describes the model, not continuous uptime.

Block	Calendar start–end	Bins	Source members	Observation period
2046	01-07–01-17	403	241,474	January
2055	04-07–04-17	181	108,008	April
2056	04-17–04-27	259	153,774	April
2058	05-07–05-17	121	72,085	May
2059	05-17–05-27	299	178,683	May
2063	06-26–07-06	527	315,603	June–July
2064	07-06–07-16	72	42,708	June–July
2067	08-05–08-15	592	355,036	August a
2068	08-15–08-25	341	203,702	August b
2069	08-25–09-04	110	66,000	August b

Table 4 | Accumulation on the actual 2026 record. Sample days sum retained members at a nominal one-second gate; span is elapsed BJT time. T and L denote the whole-template and daily-controlled lunar targets. **a,** Support and conditional independent-bin benchmark standard errors. **b,** Observed standard errors by method. "—" means fewer than three deletion groups; no missing error is assigned zero. These pointwise, nested comparisons do not have simultaneous confidence coverage or an optional-stopping guarantee.

a. Exposure, support and conditional benchmark

Sample days	Span (d)	Days / periods	T benchmark	L benchmark
2.003	2.2	3 / 1	0.362	2.985
4.007	99.9	6 / 2	0.241	2.599
8.001	131.7	12 / 3	0.143	0.911
12.005	179.6	20 / 4	0.123	0.424
16.004	217.8	28 / 5	0.107	0.314
20.105	231.3	35 / 6	0.101	0.268

b. Observed scale standard errors

Sample days	T LS	T day	T period	L LS	L day	L period
2.003	0.296	0.628	—	2.401	8.034	—
4.007	0.190	0.241	—	2.031	2.952	—
8.001	0.114	0.093	0.112	0.720	0.985	1.025
12.005	0.098	0.083	0.072	0.337	0.427	0.413
16.004	0.094	0.122	0.125	0.275	0.338	0.205
20.105	0.088	0.100	0.097	0.234	0.282	0.188

Table 5 | Joint lunar phase tests on the same observations. Six-period restricted wild tests compare the in-phase/quadrature pair with zero and with the unit, in-phase prediction. All 64 sign patterns are retained and each is studentized by its own delete-period covariance. These exploratory tests differ from fixed-phase scalar tests and the nominal delete-day ellipses in Fig. 2b. All designs use the same 2,905 observations; the primary ten-day baseline is unchanged.

Baseline	In phase	Quadrature	p, zero	p, unit in phase
5 days	0.892	-0.053	0.31250	0.90625
10 days	0.866	0.104	0.25000	0.71875
20 days	0.867	0.002	0.03125	0.62500
30 days	0.794	0.099	0.12500	0.25000
Observation period	0.754	0.067	0.03125	0.12500

Table 6 | Averaging-time support of the retained 2026 clock record. All 1,737,073 frozen members are retained before run-length requirements. Runs split at file and effective-interval boundaries or nonconsecutive nominal seconds. 4τ is the primary minimum run length; 2τ is an exploratory comparison, not a confidence level. Allan deviations are in 10^{-18} units; the last column subtracts the fixed unit prediction on identical support. Unsupported estimates are shown as —. Overlapping differences are not independent replicates. This support differs from the simultaneous B2 support in Fig. 3.

Minimum run	τ (s)	Runs / periods	Differences	Retained	Fixed tide subtracted
4τ	600	68 / 6	1,651,143	21.588	21.587
4τ	3,600	43 / 6	1,203,996	7.312	7.210
4τ	8,192	18 / 6	633,924	5.171	4.600
4τ	16,384	5 / 3	212,674	5.583	3.720
4τ	21,600	2 / 2	86,407	4.348	2.165
4τ	32,768	0 / 0	0	—	—
4τ	43,200	0 / 0	0	—	—
2τ	32,768	5 / 3	48,834	2.708	1.419
2τ	43,200	2 / 2	7	1.087	0.549

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